

BLS Performance re: 80020

Preliminary findings by Monica Gucciardi

August 2026



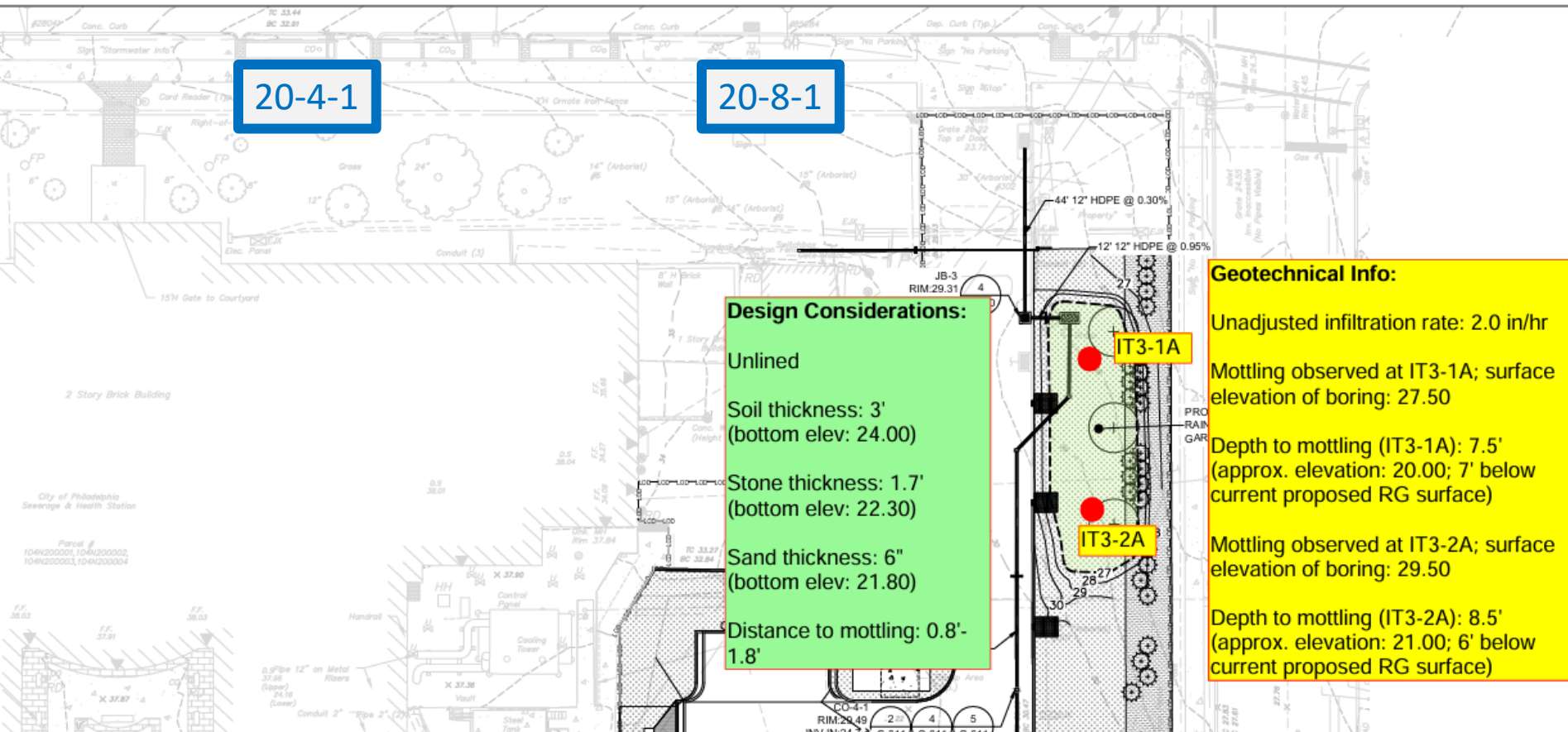
PHILADELPHIA
WATER
— DEPARTMENT —



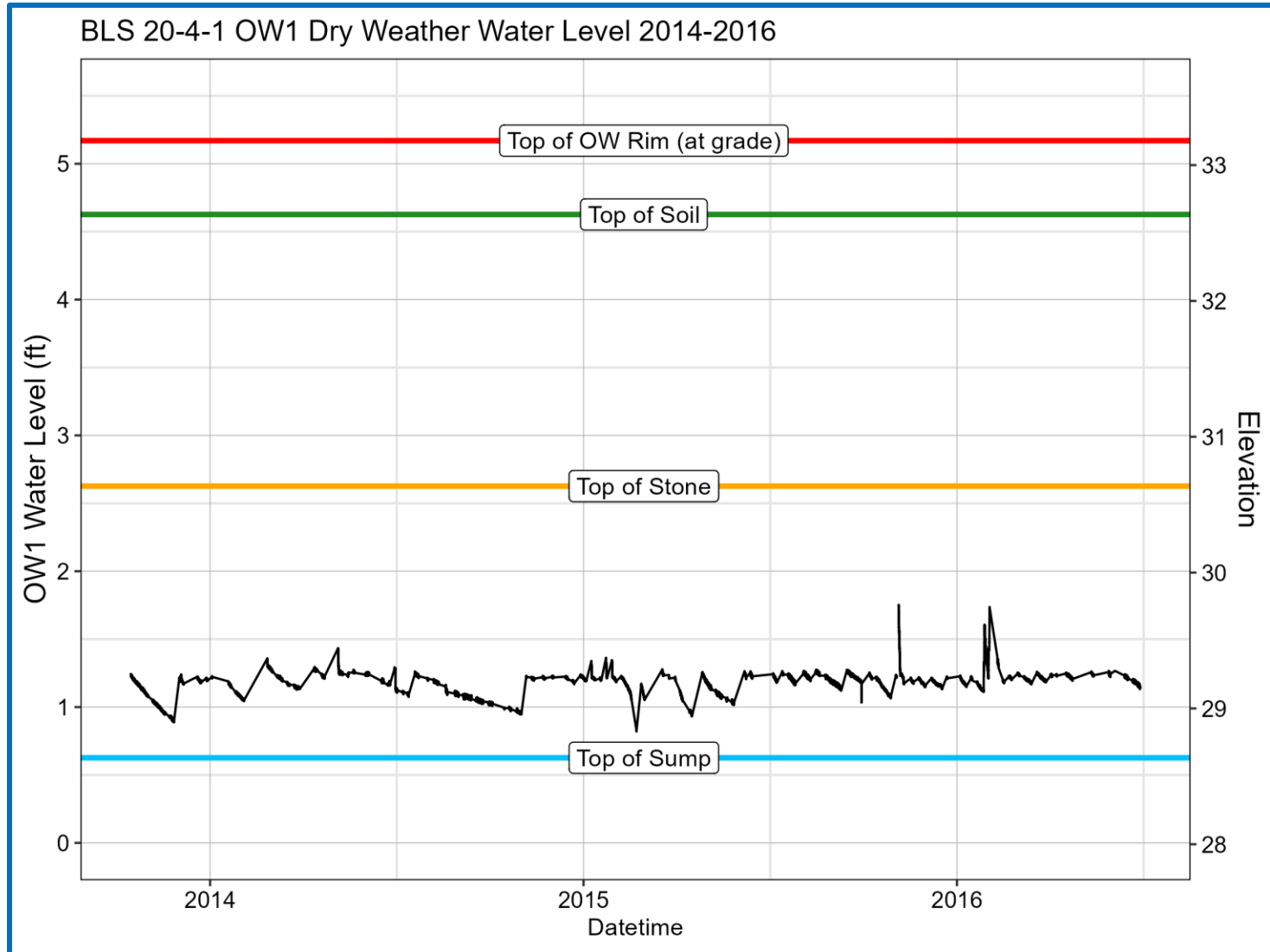
Table of Contents

- 1. Project Overview**
- 2. Water Level Plots**
- 3. Table of Elevations**
- 4. Groundwater Findings**

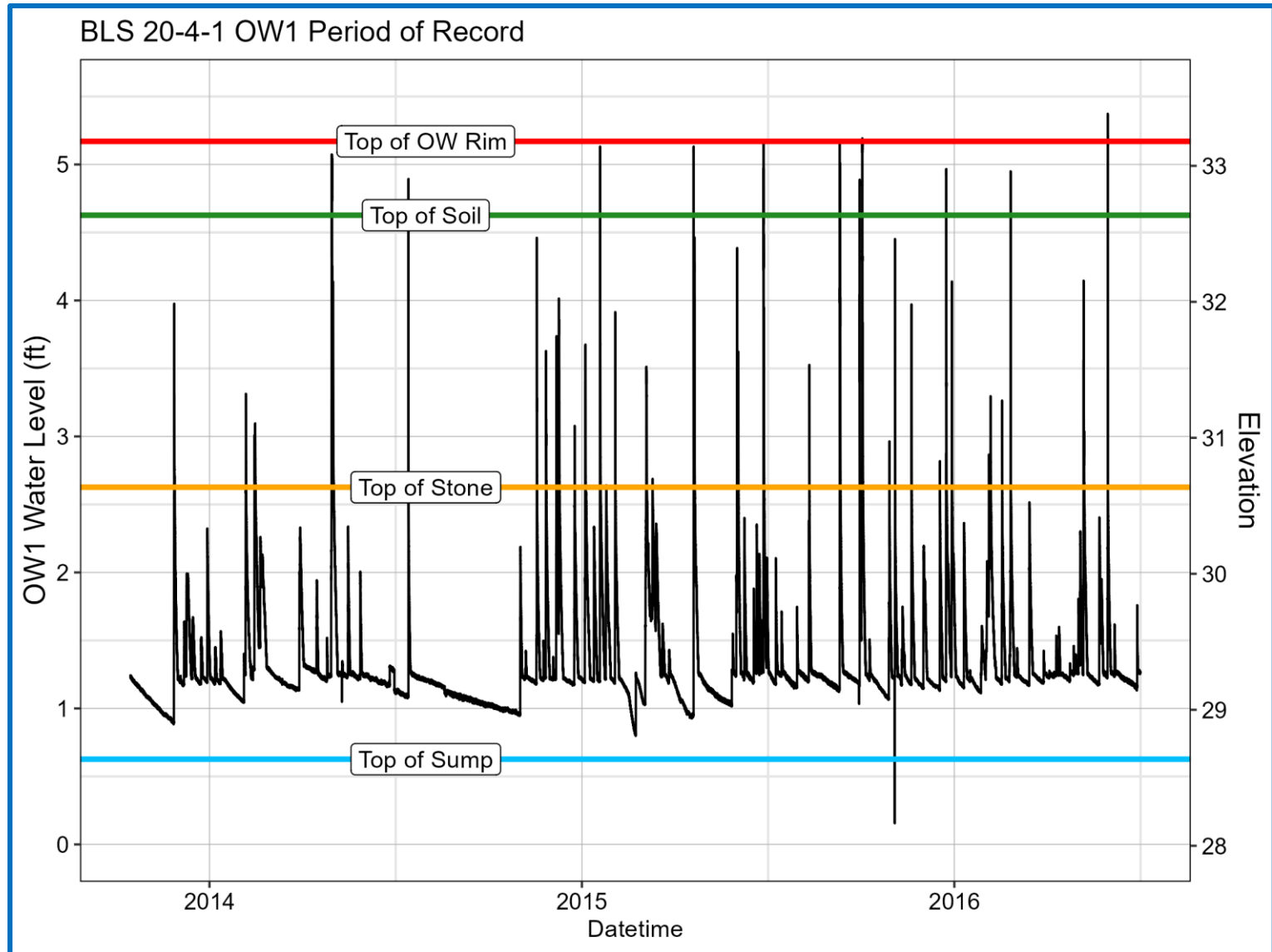
20-X GSI relative to 80020 RG3



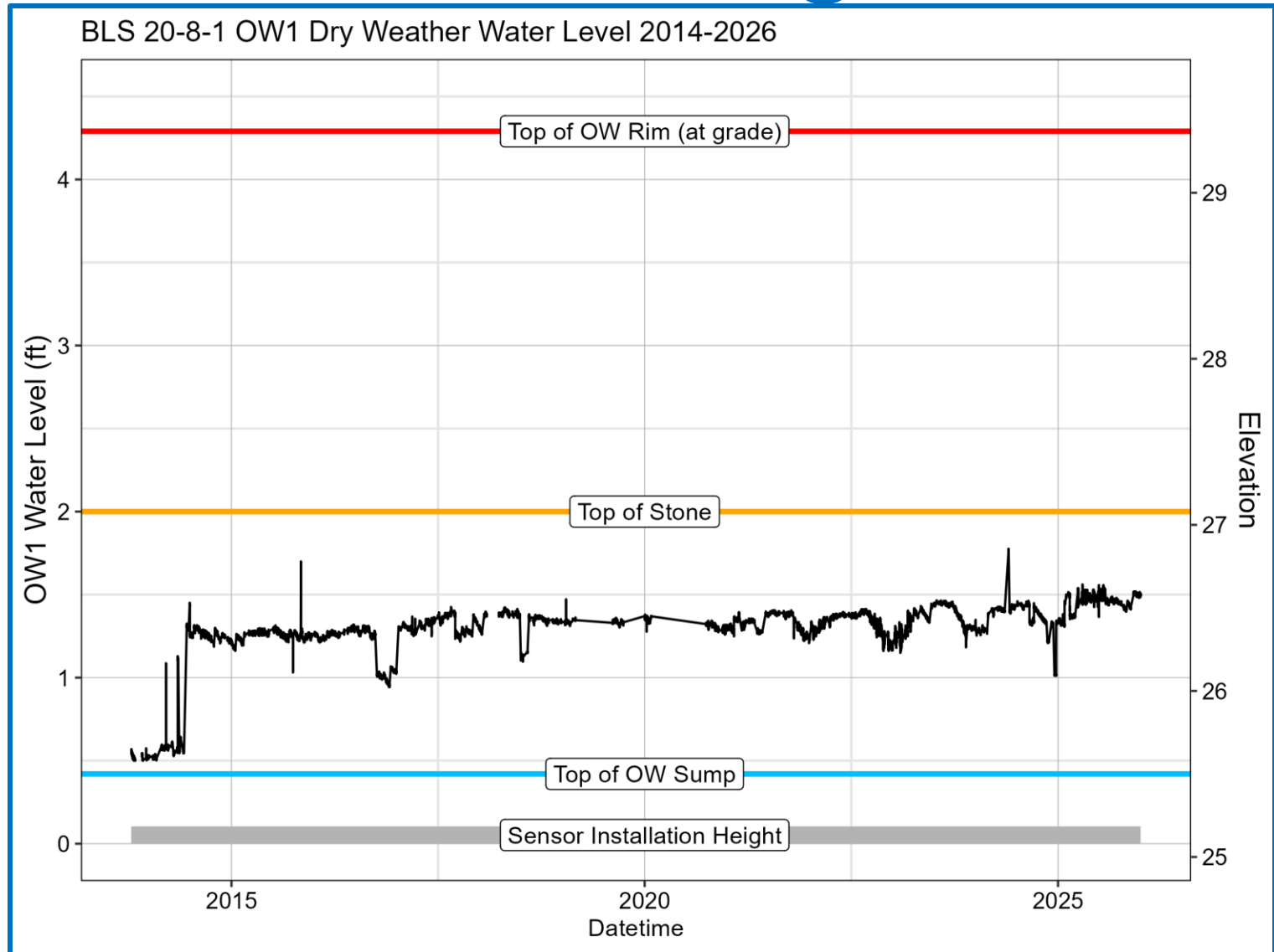
Dry Weather Standing Water 20-4



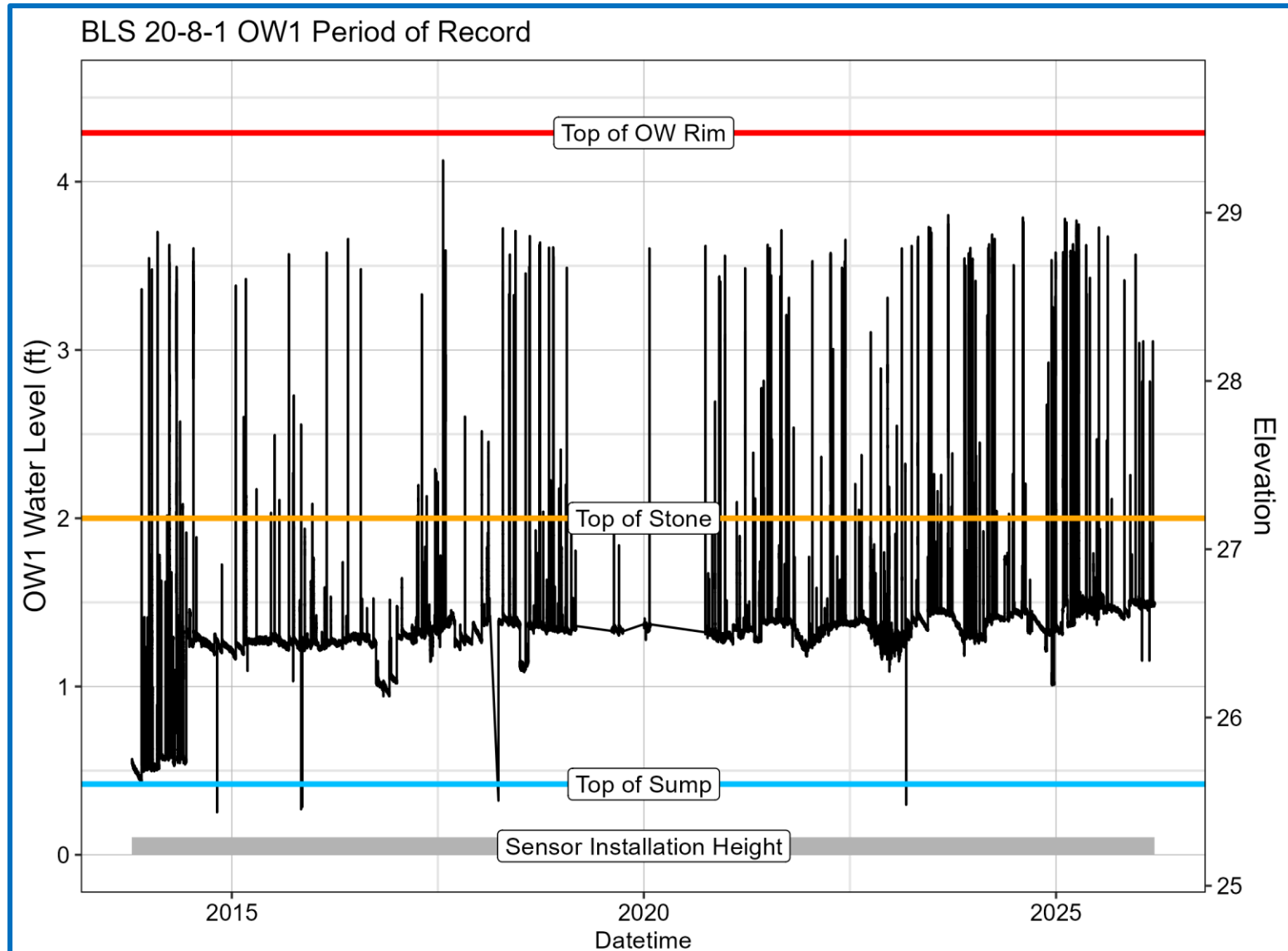
Wet Weather Time Series 20-4



Dry Weather Standing Water 20-8



Wet Weather Time Series 20-8



Strata Diagram for 80020 site

August 2026 MARS Subsurface GSI Performance Eval for 80020 Design Phase

Elevation	80020 proposed RG1 elevations	80020 proposed RG2 elevations	80020 proposed RG3 elevations	Standing water observed @ 20-4-1	Standing water observed @ 20-8-1
Phila. datum	Mottling detected in RG1 geotech log	Mottling detected in RG2 geotech log	Mottling detected in RG3 geotech log	Rim elevation measured via MARS field survey	Rim elevation measured via MARS field survey
33	Proposed bottom of RG1 soil [33.00]			20-4-1 OW rim elevation [33.18]	
32.5					
32					
31.5					
31	Proposed bottom of RG1 stone [31.30]				
30.5					
30					
29.5				Top of 20-4-1 dry weather standing water range [~29.75]	
29				Bottom of 20-4-1 dry weather standing water range [~29]	20-8-1 OW rim elevation [29.37]
28.5				20-4-1 Dry Weather Standing Water Range (2014-2016)	
28		Proposed bottom of RG2 soil [28.00]		Standing water is approx. 3.5 feet below grade	
27.5					
27					
26.5		Proposed bottom of unlined RG2 stone [26.30]			Top of 20-8-1 dry weather standing water range [~26.75]
26		Proposed bottom of lined RG2 stone [25.80]			Bottom of dry weather standing water level [~26.25]
25.5	Mottling detected @ IT1-1A [25.80]	Mottling detected @ IT2-2A [25.50]			20-8-1 Dry Weather Standing Water Range (2014-2026)
25	IT1-1A boring @ 37.50, 11.7 feet above mottling	IT2-2A boring @ 32.50, 7 feet above mottling			Standing water is approx. 3 feet below grade
24.5					
24			Proposed bottom of RG3 soil [24.00]		
23.5					
23					
22.5					
22			Proposed bottom of RG3 stone [22.30]		
21.5					
21			Mottling detected @ IT3-2A [21.00]		
20.5			IT3-2A boring @ 29.50, 8.5 feet above mottling		
20			Mottling detected @ IT3-1A [20.00]		
19.5		Mottling detected @ IT2-1A [19.50]	IT3-1A boring @ 27.50, 7.5 feet above mottling		
19		IT2-1A boring @ 31.50, 12 feet above mottling			
18.5					
18					

```

1 with controlwells as (select * from fieldwork.tbl_ow where ow_suffix like 'CW%'),
2
3 gwdates as (select ow_uid, max(dtime) from data.tbl_gw_depthdata_raw g
4             where g.ow_uid in (select ow_uid from controlwells) group by ow_uid)
5
6 select loc.smp_id as "SMP ID",
7        loc.lon_wgs84 as "Longitude",
8        loc.lat_wgs84 as "Latitude",
9        2 * 6371 * 0.6214 * asin(sqrt(
10          sin((radians(lat_wgs84) - radians(40.01))/2)^2 + -- Haversine formula for spherical distance
11          cos(radians(40.01)) * cos(radians(lat_wgs84)) * -- Source: omnicalculator.com
12          sin((radians(lon_wgs84) + radians(75.0967))/2)^2)) as "Miles from BLS (approx)",
13        g.max as "Latest CW Record"
14 from admin.tbl_smp_loc loc
15      left join controlwells c on loc.smp_id = c.smp_id
16      left join gwdates g on g.ow_uid = c.ow_uid
17 where c.smp_id is not null or
18        loc.smp_id = '20-8-1'
19 order by "Miles from BLS (approx)" asc

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Most relevant groundwater well to analyze is 274-4-1 CW1 at Roosevelt Playground, approximately 2.5 miles from

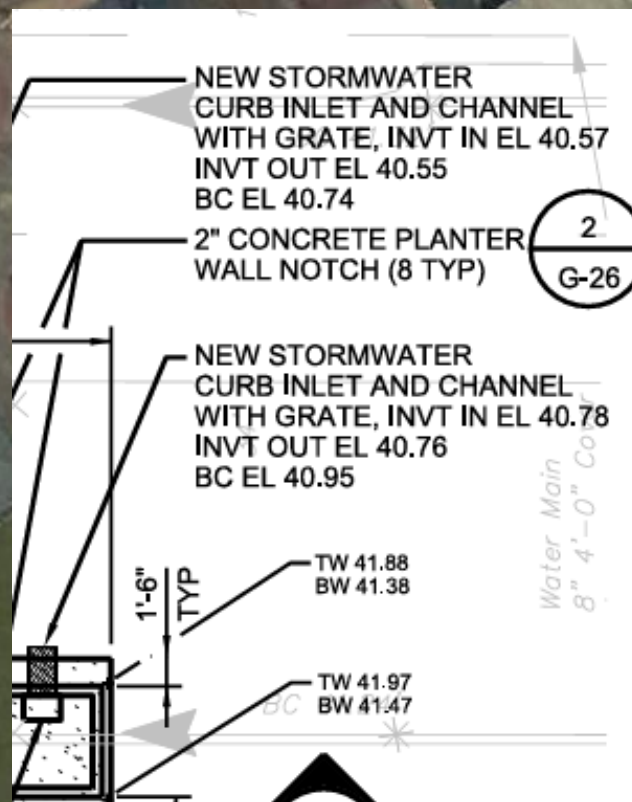
BLS

Data Output Messages Notifications



Showing rows: 1 to

	SMP ID text	Longitude double precision	Latitude double precision	Miles from BLS (approx) double precision	Latest CW Record timestamp with time zone
1	20-8-1	-75.0967	40.01	0	[null]
2	271-1-1	-75.0689	39.9995	1.6405267999616615	2019-02-21 10:15:00-05
3	274-4-1	-75.0533	40.026	2.548843017162637	2026-06-26 13:00:00-04
4	285-1-1	-75.163	40.0094	3.5090722755950616	2019-05-14 15:00:00-04
5	403-2-1	-75.1543	39.9367	5.912236693371212	2017-04-01 00:55:00-04



Approx location of closest Roosevelt planter box

Roosevelt Playground control well approximate elevation 40.5

Roosevelt Playground 274-4-1 CW1 Groundwater Elevation

